Chapter-2: Wave & Oscillation

Course code: PHY-101
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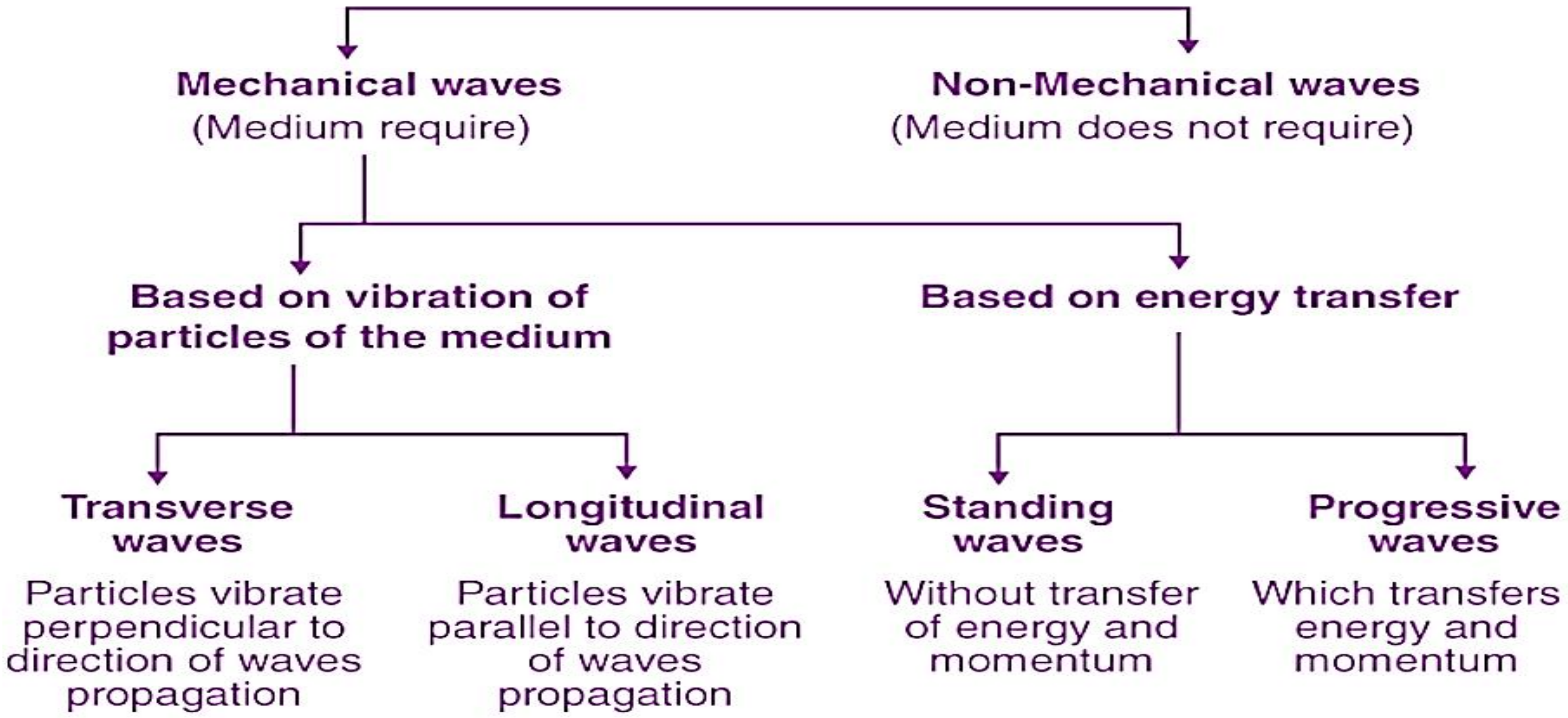
Wave & Oscillation

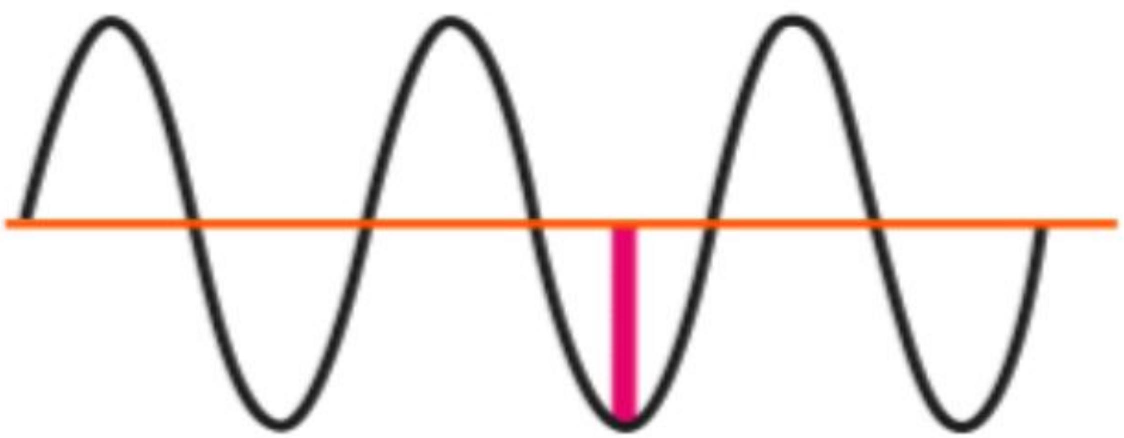
A **wave** is a **disturbance or vibration** that travels through space or a medium, carrying **energy** from one point to another without transporting **matter or particle** permanently.

Classification of waves:

WAVE



Transverse & Longitudinal Wave

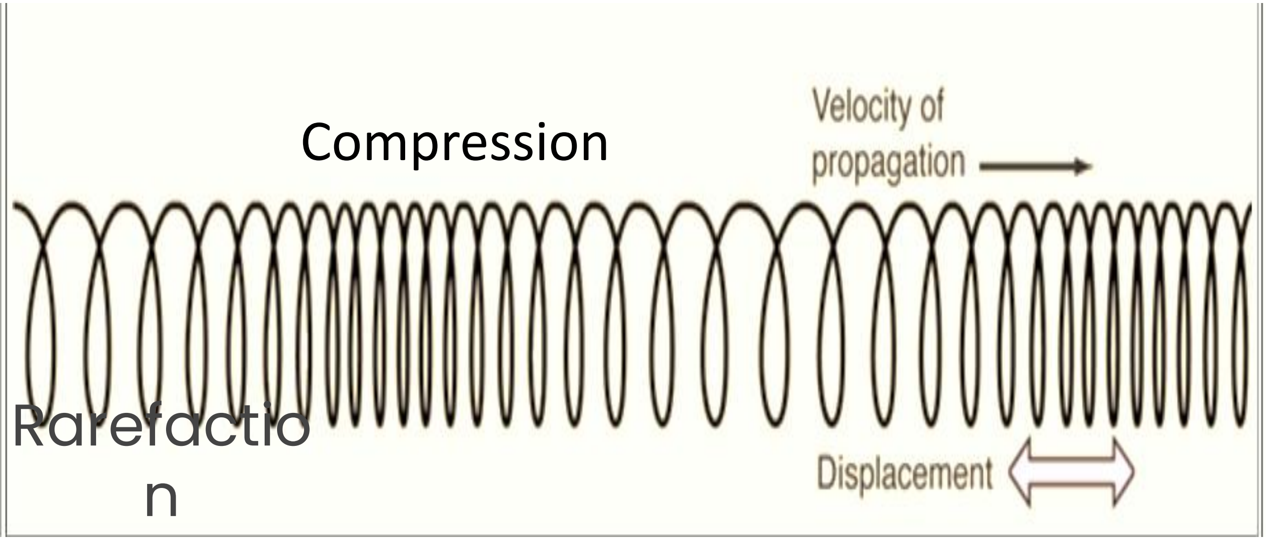


Transverse Wave

The displacement of the medium is **perpendicular** to the direction of propagation of the wave.

i) A [ripple on a pond]
ii) A [wave on a string]

https://youtu.be/Voki54XpVEA?si=-omkFt2YN1zTUqvz



Longitudinal Wave

The displacement of the medium is **parallel** to the propagation of the wave.

i) Sound waves in air.

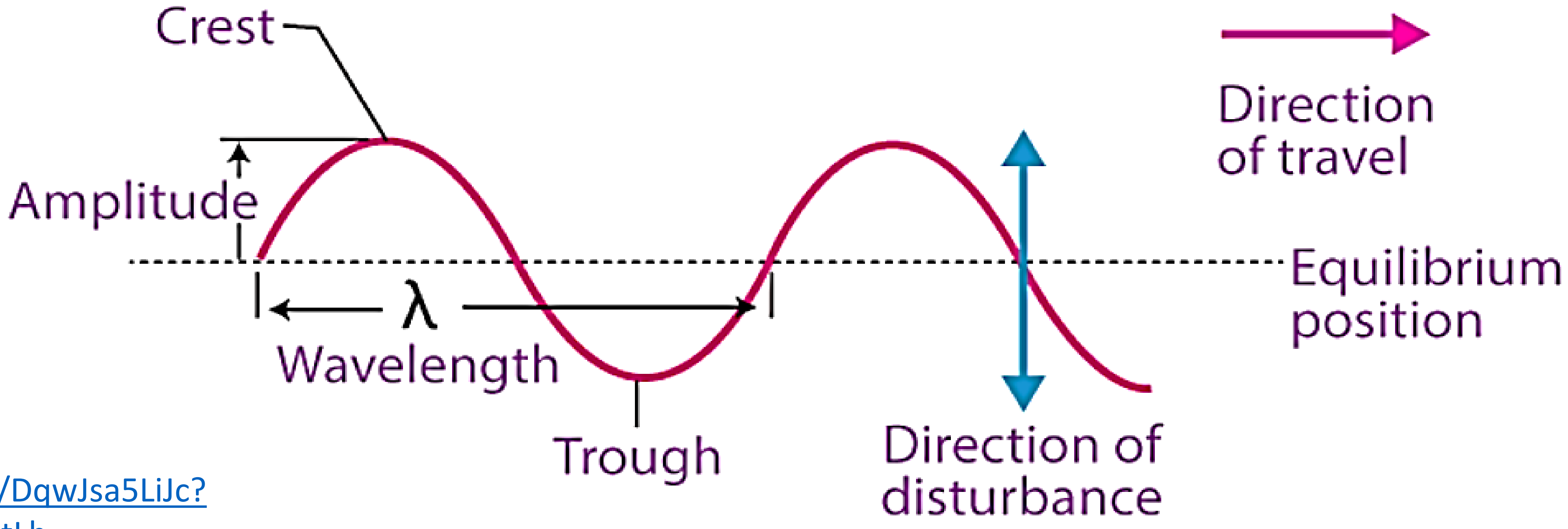
https://youtu.be/pHgTOKxNWqc?si=kkgtTxji9cdQb4XS

Progressive Wave

A **progressive wave** is a wave in which the **disturbance travels** through a medium, **transferring energy** from one particle to another without permanently displacing of matter. Each particle of the medium vibrates with the **same amplitude** but **at different phases.**

Example:

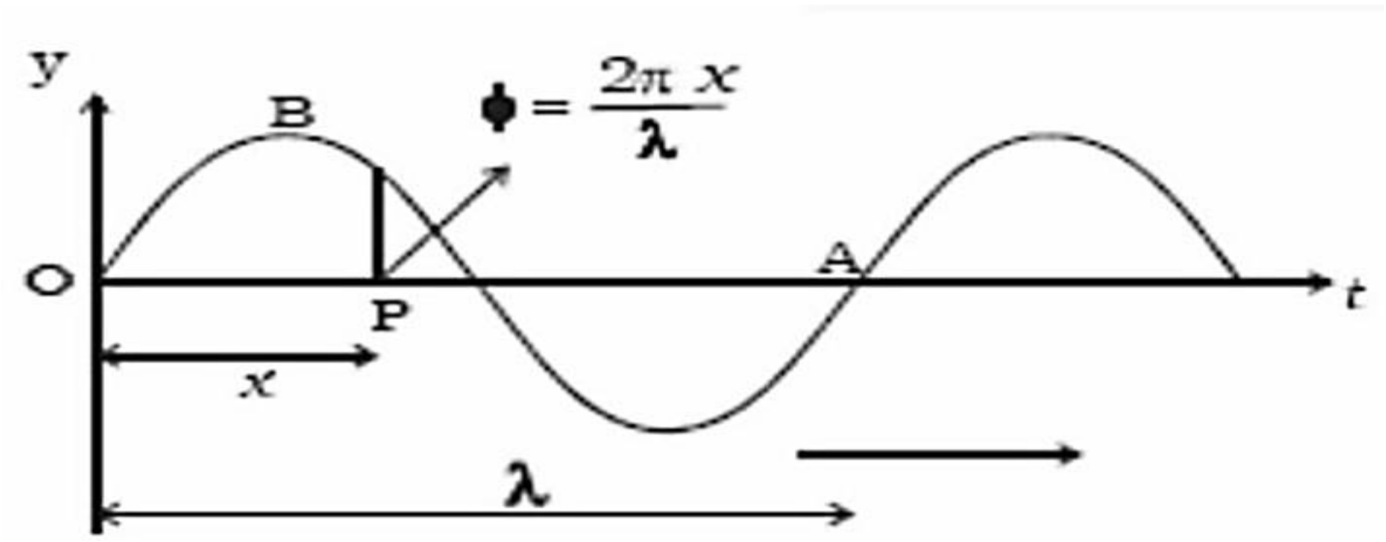
i) ripples moving outward on water. Ii) sound traveling in air.



https://youtu.be/DqwJsa5LiJc?si=VJ9af0B22nXbtLh-

Figure: Plane Progressive Wave

Plane Progressive Wave



We know the equation of displacement of a particle forming simple harmonic motion is given by,

$$y = \mathrm{A}\sin\ (\omega t + \ \varphi)$$

Now, at time t = 0 second, the initial phase at position O is $\varphi = 0$ degree, then displacement equation of a particle at position O is given by, $y = \mathrm{A}\sin\ (\omega t + \ \varphi)$

$$y = \mathrm{A}\sin\ (\omega t + 0)$$

$$y = \mathrm{A}\sin\ (\omega t) \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots(1)$$

Plane Progressive Wave

Displacement equation of another particle at point P is given by,

$y = \mathrm{A}\sin\ (\omega t - \varphi)$ …………………………………..(2)

For the wavelength of λ, the phase difference = 2π

For unit length, the phase difference = $\frac{2\pi}{\lambda}$

If the wave travels of x distance, the phase difference will be = $\frac{2\pi}{\lambda} \times x = \frac{2\pi x}{\lambda}$

Here, Phase difference $(\Delta\varphi) = \frac{2\pi}{\lambda} \times Path\ Difference\ (\Delta x)$

Now, from equation (2), we get, $y = \mathrm{A}\sin\ (\omega t - \frac{2\pi x}{\lambda})$

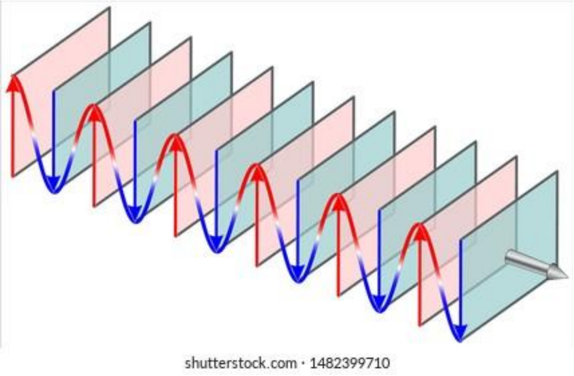
$= \mathrm{A}\sin\ (\omega t - kx\ )$ ……………………………………………….(3)

where, k is called the wave number, k = $\frac{2\pi x}{\lambda}$

But $\omega = \frac{2\pi}{T}$, so from (3) we get, $\mathrm{y} = \mathrm{A}\sin\left(\frac{2\pi}{T}\mathrm{t} - \frac{2\pi}{\lambda}x\right) = \mathrm{A}\sin 2\pi\left(\frac{t}{T} - \frac{x}{\lambda}\right)$

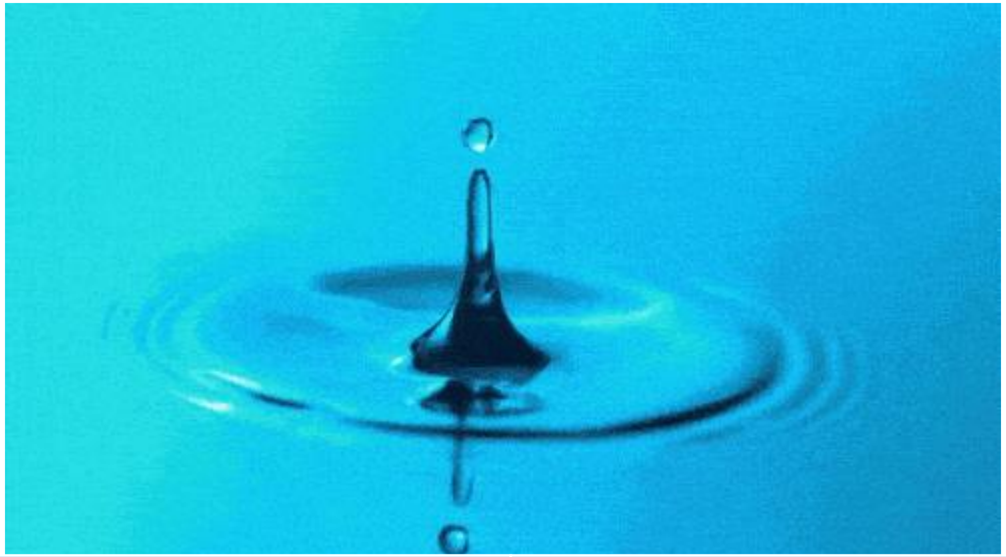
Again we know, $\omega = \frac{2\pi}{T} = 2\pi f = \frac{2\pi v}{\lambda}$, So from equation (3), we can write

$$\mathrm{y} = \mathrm{A}\sin\left(\frac{2\pi v}{\lambda}\mathrm{t} - \frac{2\pi}{\lambda}x\right) = \mathrm{A}\sin\frac{2\pi}{\lambda}(vt - x\ )$$



https://youtu.be/zUBXiTkrmSo?si=jvf8d5iZaftavi4Q

Difference Between Progressive & Plane Wave

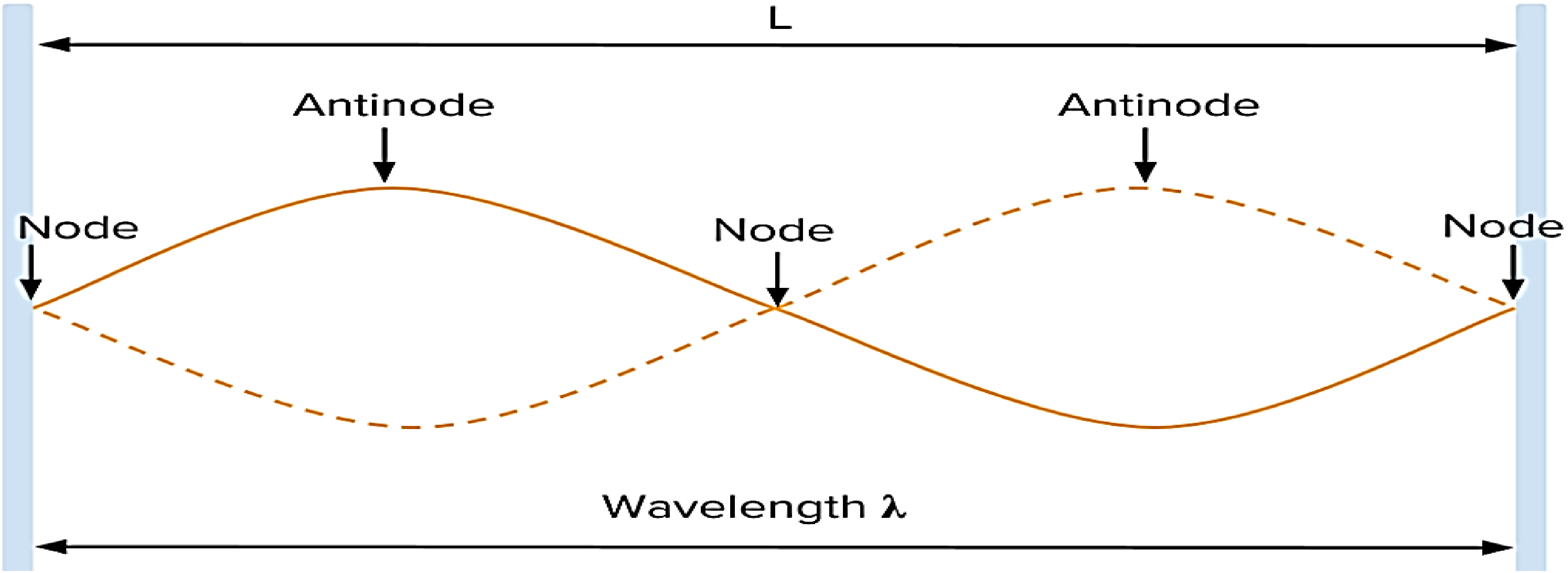


https://tenor.com/search/ripple-in-the-water-gifs

Progressive Wave	Plane Wave
It travels continuously through a medium, transferring energy from one point to another.	It represents an idealized wave where wavefronts (surfaces of equal phase) are parallel planes.
The wavefronts can be **spherical** or **cylindrical**, depending on the source.	The wavefronts are **flat and parallel**, assuming the source is very far away.
Ripples spreading on the surface of water when a stone is dropped.	Light waves coming from the Sun reaching Earth — wavefronts are almost plane because the Sun is far away.

Standing / Stationary Wave

A **standing wave** is a wave formed by the superposition of two identical progressive waves traveling in **opposite directions**, resulting in a wave pattern with fixed **nodes (zero displacement)** and **antinodes (maximum displacement),** where energy does not propagate forward. Example: Vibrations on a **guitar or violin string, Microwave oven cavity.**



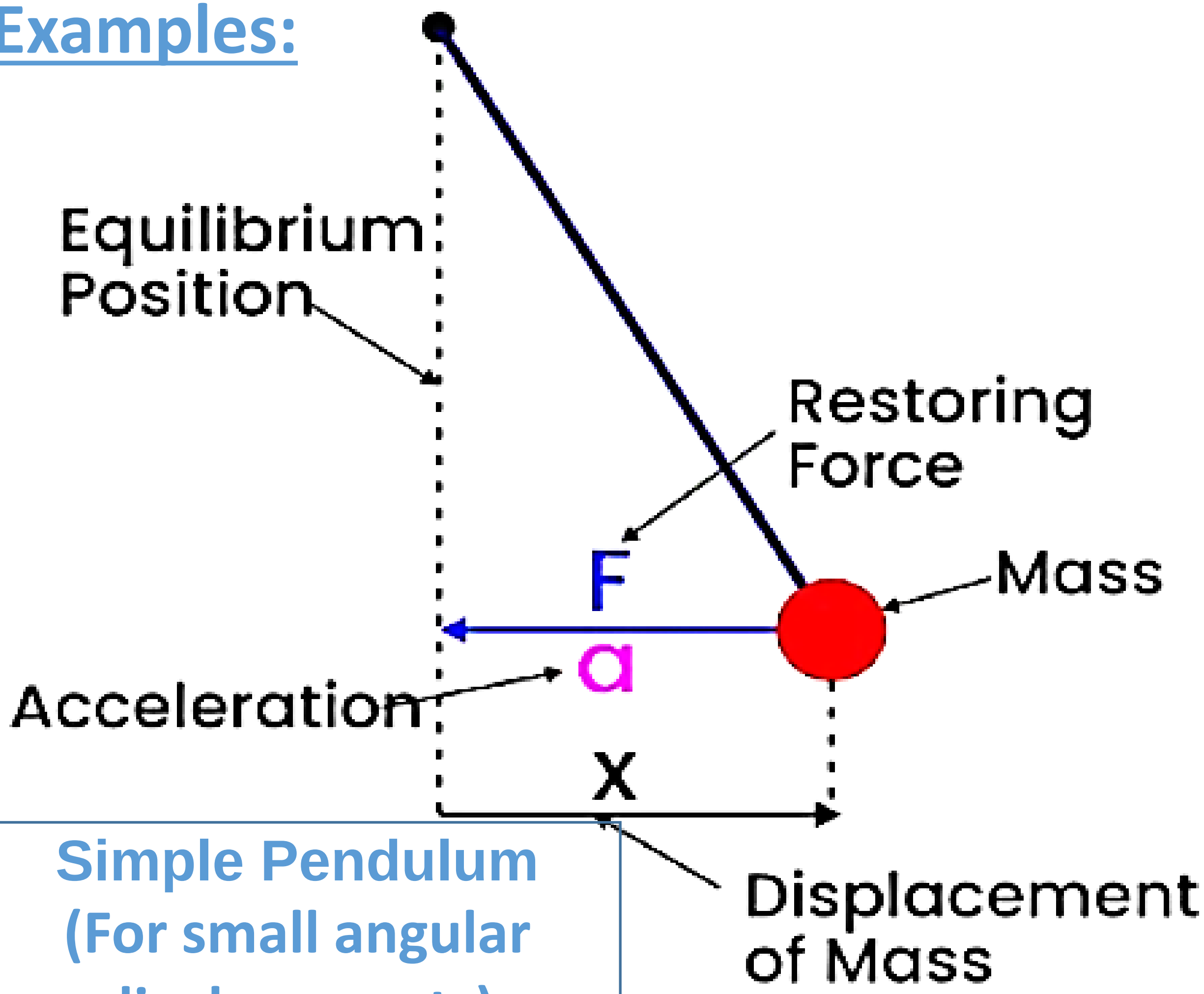
Simple Harmonic Motion

1) Acceleration & Restoring Force acts in the same direction
2) Displacement acts in the opposite direction

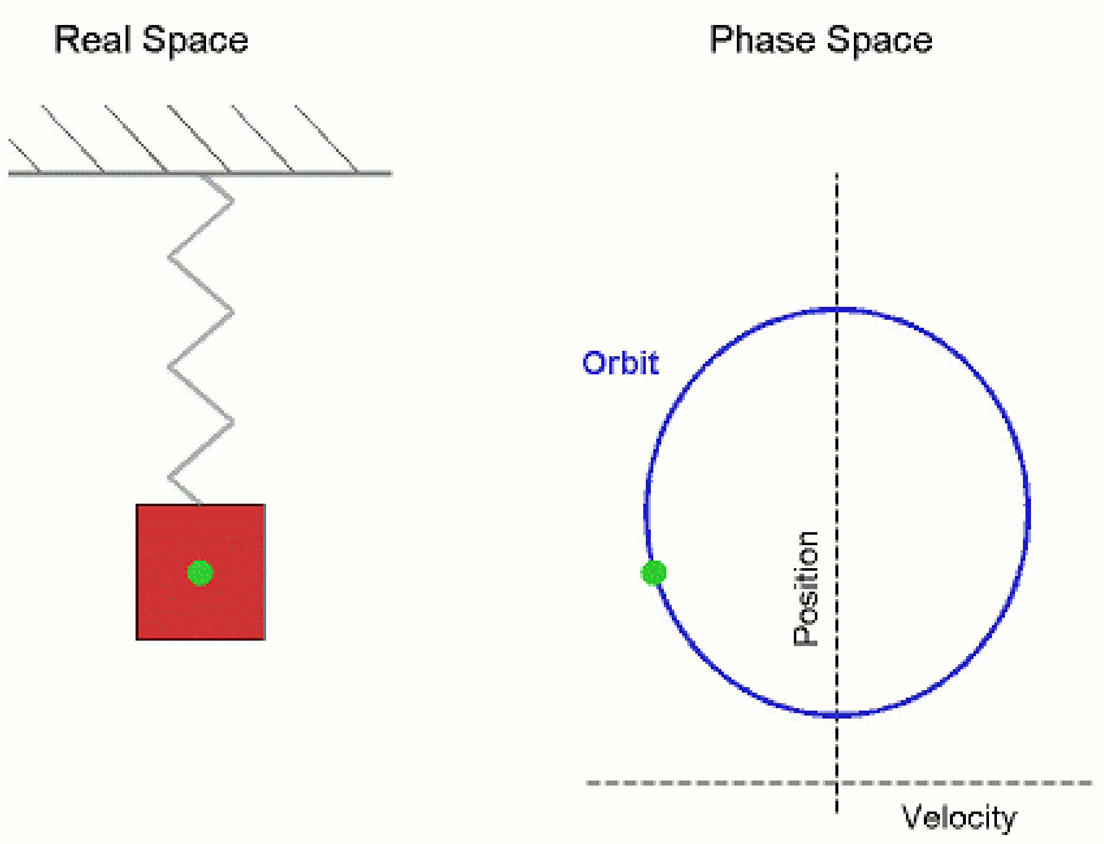
Here, $F \propto -x$

$a \propto -x$

Examples:



Simple Pendulum (For small angular displacements)



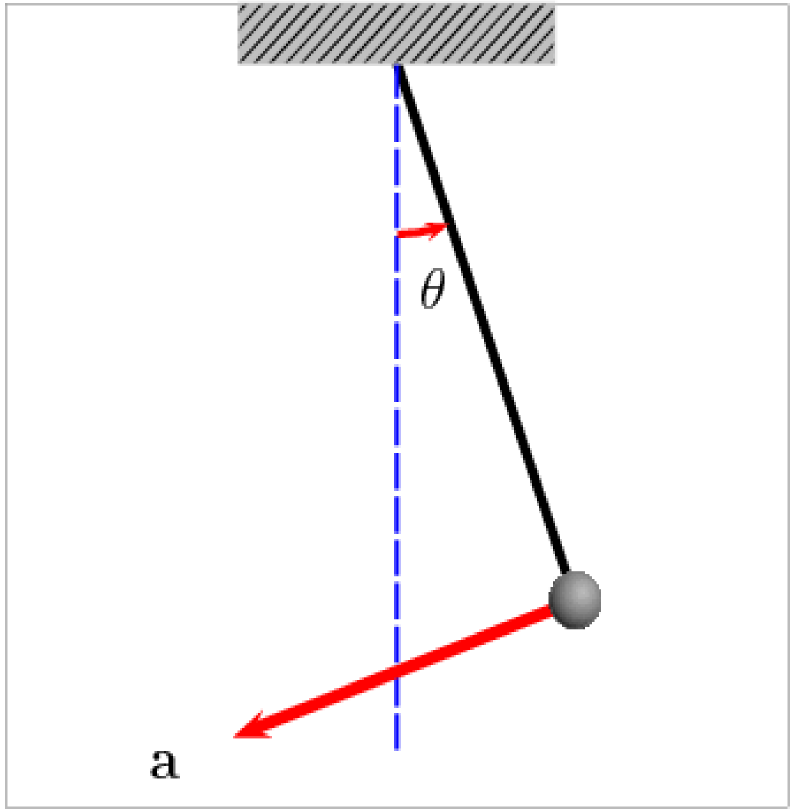
Spring (Perfect Example)
SHM

Simple Harmonic Motion

Simple harmonic motion is a type of **periodic motion** in which a particle moves to and fro about **a mean (equilibrium) position** under a **restoring force** that is directly **proportional to its displacement** and always directed towards the mean position. SHM

The **main properties of Simple Harmonic Motion (SHM):**

1) The motion is **periodic**.
2) The particle oscillates **about a mean (equilibrium) position**.
3) The **restoring force or acceleration** is always directed towards the **mean position**.
4) The **magnitude of restoring force (or acceleration)** is **directly proportional to the displacement** from the mean position.



So, $F \propto -x \quad and$

Differential Equation of SHM

In SHM, the restoring force acting on the particle is always proportional to the displacement from the mean position and directed towards it. **Here, $F \propto -x$**

Hook's law: $F = -kx$ .... **(1)**

where,

k = force constant (spring constant in spring–mass system),

x = displacement from equilibrium.

According to Newton's second law,

F= ma .... **(2)**

or, ma = $-kx$ **(Equating (1) and (2), we get)**

or, $m\frac{d^2x}{dt^2} = -kx \left[\because a = \frac{d^2x}{dt^2}\right]$

or, $m\frac{d^2x}{dt^2} + kx = 0$

or, $\frac{d^2x}{dt^2} + \frac{k}{m}x = 0$

or, $\frac{d^2x}{dt^2} + \omega^2 x = 0$ ....(3) $\left[\because \omega^2 = \frac{k}{m}\right]$

Equation (3) is known as the 2nd order differential equation of Simple Harmonic Motion. The solution of equation (3) is given by $\mathbf{X = A\sin(\omega t + \varphi)}$.

Here, $\omega^2 = \frac{k}{m}$

$\therefore \omega = \sqrt{\frac{k}{m}}$, here,

$\boldsymbol{\omega}$ ***is known as the angular frequency of Simple harmonic motion.***

Velocity & Acceleration in SHM

The velocity of the vibrating particle in SHM is given by;

$$v = \frac{dx}{dt} = \frac{d}{dt}\,[A\sin(\omega t + \varphi)]$$

$$= \omega A\cos(\omega t + \varphi)$$

$$= \omega\sqrt{A^2 cos^2(\omega t + \varphi)}$$

$$= \omega\sqrt{A^2[1 - sin^2(\omega t + \varphi)]}$$

$$= \omega\sqrt{A^2 - A^2 sin^2(\omega t + \varphi)]}$$

$$v = \omega\sqrt{A^2 - x^2}$$

At the mean position $(x = 0)$, the velocity is maximum,

$$v_{max} = \omega\sqrt{A^2 - x^2} = \omega\sqrt{A^2 - 0^2} = \omega\sqrt{A^2} = \omega A$$

$$\therefore\ \boldsymbol{v_{max} = \omega A}$$

The acceleration of the vibrating particle in SHM is given by;

$$a = \frac{dv}{dt}$$

$$= \frac{d^2x}{dt^2}$$

$$= \frac{d^2}{dt^2}\,[A\sin(\omega t + \varphi)]$$

$$= -\omega^2 A\sin(\omega t + \varphi)$$

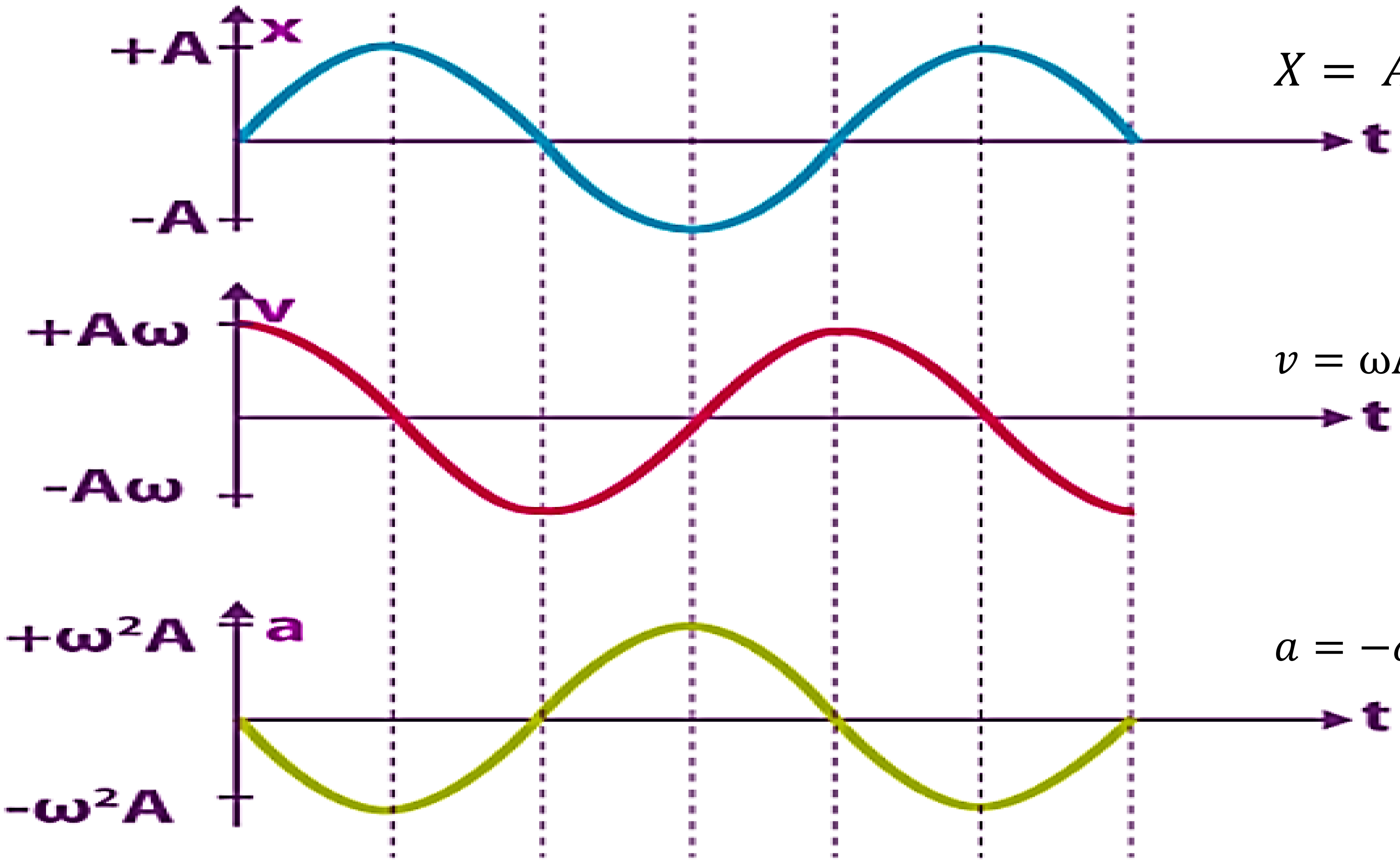
$$a = -\omega^2 x$$

When the particle under simple harmonic motion reaches the maximum position, i.e.

$x = A, the\ acceleration\ is\ maximum.$

$$\boldsymbol{a_{max} = -\omega^2 A}$$

Velocity & Acceleration Graphs In SHM



$$X = A\sin(\omega t + \varphi)$$

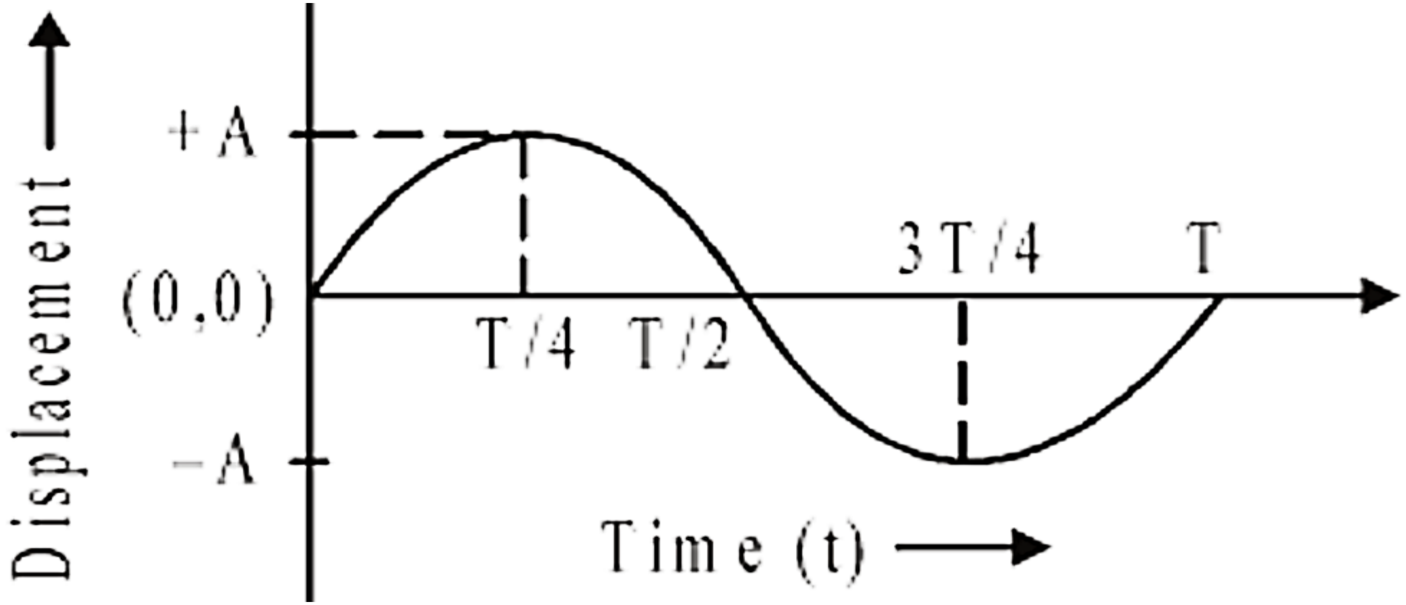
$$v = \omega A\cos(\omega t + \varphi)$$

$$a = -\omega^2 A\sin(\omega t + \varphi)$$

Different Parameters of SHM

The solution of equation (3) is given by $X = A \sin(\omega t + \varphi)$. **X** is the **instantaneous displacement** of the oscillating particle from its **mean (equilibrium) position** at any time t.

1) **Amplitude (A)** is the **maximum displacement** of an oscillatory particle from its mean (equilibrium) position.
2) **Angular frequency (ω)** is the rate at which a particle oscillates in simple harmonic motion, measured in **radians per second (rad/s)**.

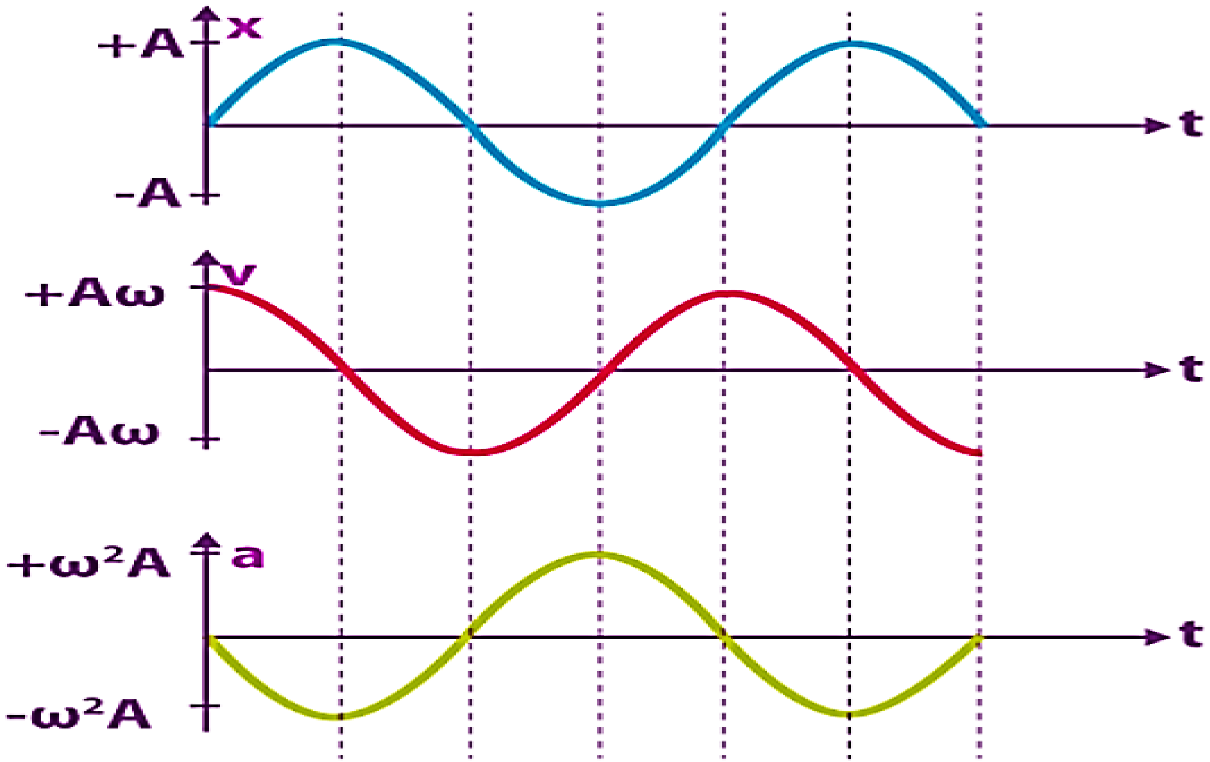


3) **The time** taken for a complete oscillation by a particle executing simple harmonic motion is called time period (**T**). The **frequency** of simple harmonic motion is the **number of complete oscillations made by the particle in one second**.

Different Parameters of SHM

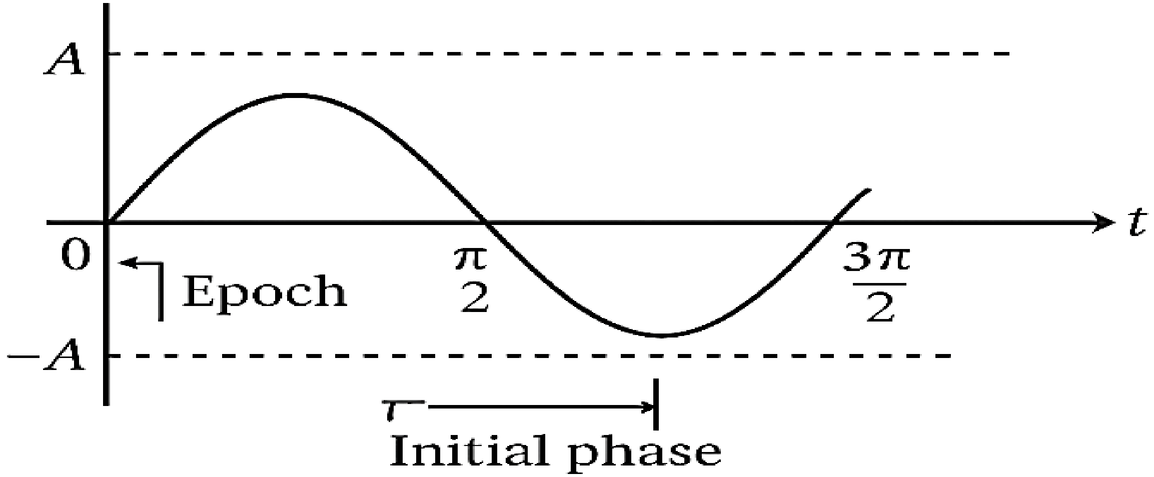
Phase ($\omega t + \varphi$):

The state of motion of a particle at a particular time (any time t) is called its phase. The state of motion means the displacement, velocity, acceleration etc.



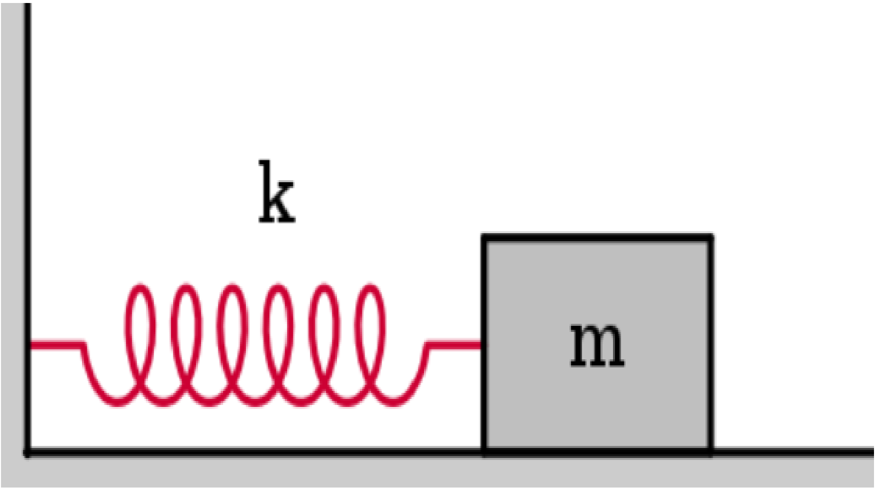
Epoch or initial phase (φ_0):

In simple harmonic motion (SHM), the epoch, also known as the initial phase, is the phase of a vibrating particle at time t = 0. It is a constant value, typically denoted by φ_0, and it describes the particle's initial position and direction of motion when the motion begins.



Equation of Time Period, Frequency of SHM

Relation of time period with angular frequency: $T = \frac{2\pi}{\omega}$



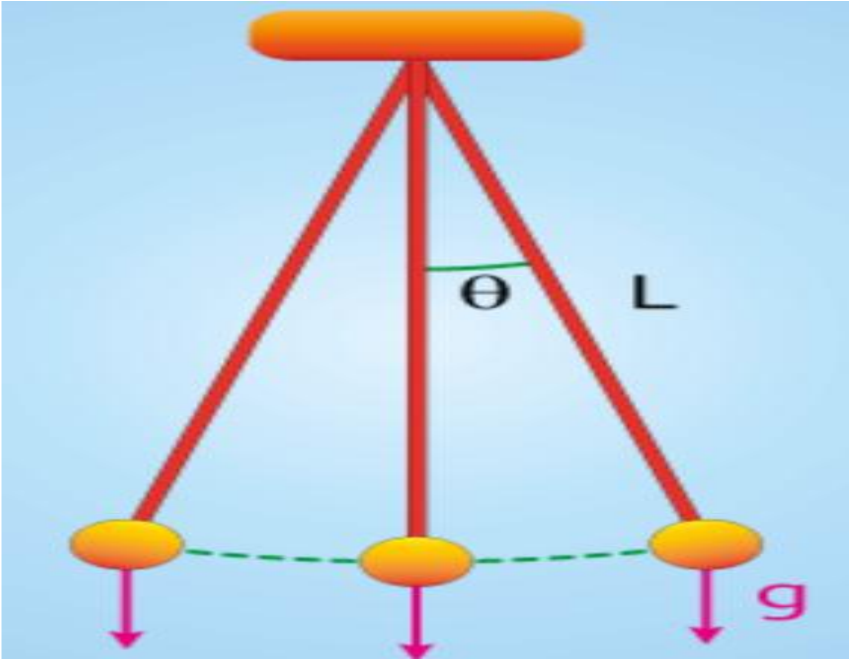
Spring–mass system: $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{k}{m}}}$ $\left[\because \omega = \sqrt{\frac{k}{m}}\right]$

$$\therefore \quad T = 2\pi\sqrt{\frac{m}{k}}$$

$$\therefore \quad f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

Simple pendulum, small oscillations : For pendulum length L, under gravity g: $T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{g}{L}}}$ $\left[\because \omega = \sqrt{\frac{g}{L}}\right]$

$$\therefore T = 2\pi\sqrt{\frac{L}{g}}$$



Relation of frequency with time period: $f = \frac{1}{T}$

Kinetic & Potential Energy In SHM

Kinetic Energy Calculation:

In SHM, The displacement of the particle at any time, t:

$X = A \sin(\omega t + \varphi)$.

Velocity, $v = \frac{dx}{dt} = \omega A \cos(\omega t + \varphi)$.

Kinetic Energy, K.E. $= \frac{1}{2} m v^2$

$$= \frac{1}{2} m \times \{\omega A \cos(\omega t + \varphi)\}^2$$

$$= \frac{1}{2} m\omega^2 A^2 \cos^2(\omega t + \varphi)$$

Potential Energy (PE) Calculations:

Restoring force: F = −kx, where, k = $m\omega^2$

So, Potential Energy = $\frac{1}{2} kx^2$

$$= \frac{1}{2} m\omega^2 \times \{A \sin(\omega t + \varphi)\}^2$$

$$= \frac{1}{2} m\omega^2 A^2 \sin^2(\omega t + \varphi)$$

Total Energy In SHM

Total Energy: E = Kinetic Energy (K. E.) + Potential Energy (P. E.)

$$= \frac{1}{2}mv^2 + \frac{1}{2}kx^2$$

$$= \frac{1}{2}m\omega^2A^2cos^2(\omega t + \varphi) + \frac{1}{2}m\omega^2A^2sin^2(\omega t + \varphi)$$

$$= \frac{1}{2}m\omega^2A^2\{cos^2(\omega t + \varphi) + sin^2(\omega t + \varphi)\}$$

$$= \frac{1}{2}m\omega^2A^2 \text{ .... (4)}$$

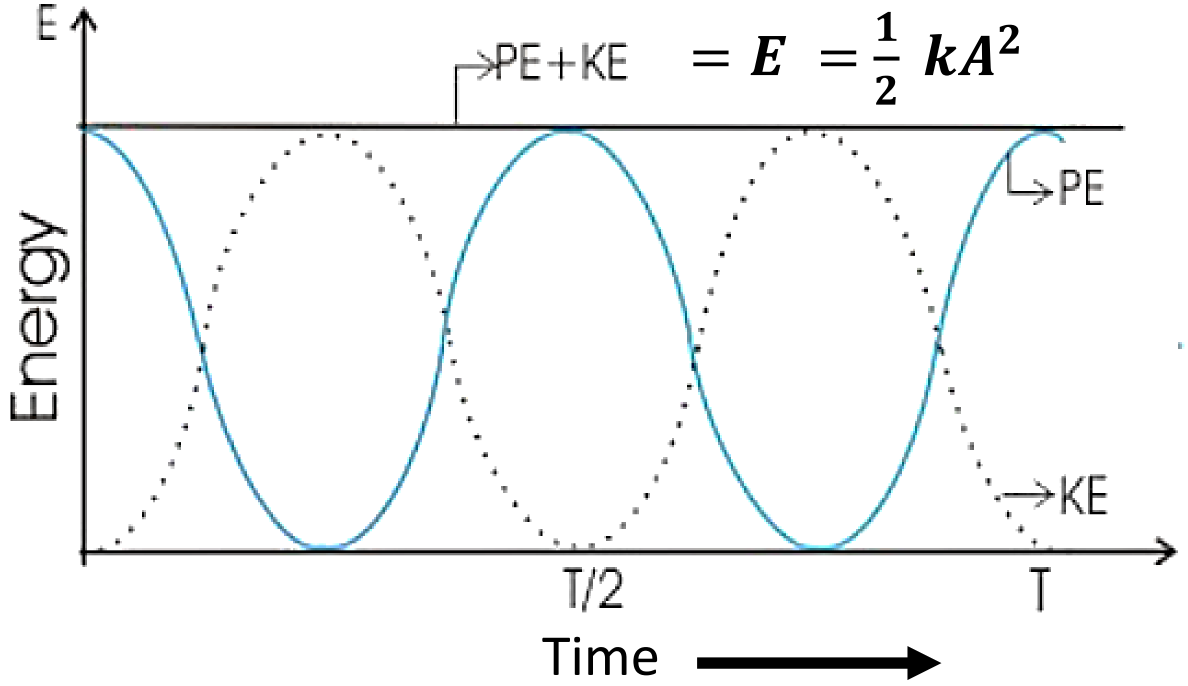
$$= \frac{1}{2}m \times \frac{k}{m} \times A^2 \qquad \left[\because \omega^2 = \frac{k}{m}\right]$$

$$\therefore E = \frac{1}{2}kA^2 \text{ .... (5)}$$

$$\therefore E \propto A^2 \text{ [Since, } \frac{1}{2}k \text{ is constant]}$$

Here, from equation (5), total energy of the particle executing simple harmonic motion is directly proportional to the squared of amplitude (A) and force constant (k) of the wave. Also, total energy is independent of time. So, at any time t, total energy is constant.

Energy Related Graphs In SHM



i) At $n \times \frac{T}{4}$ (n = 0,2,4,… ,even), the P.E. is maximum and K.E. is minimum.

ii) At (n+1) × $\frac{T}{4}$ $(n = 0,2,4, \dots \text{even})$, the K.E. is maximum and P.E. is minimum.

iii) At (n+1) × $\frac{T}{8}$ $(n = 0,2,4, \dots \text{even})$, the K.E. and P.E. equal.

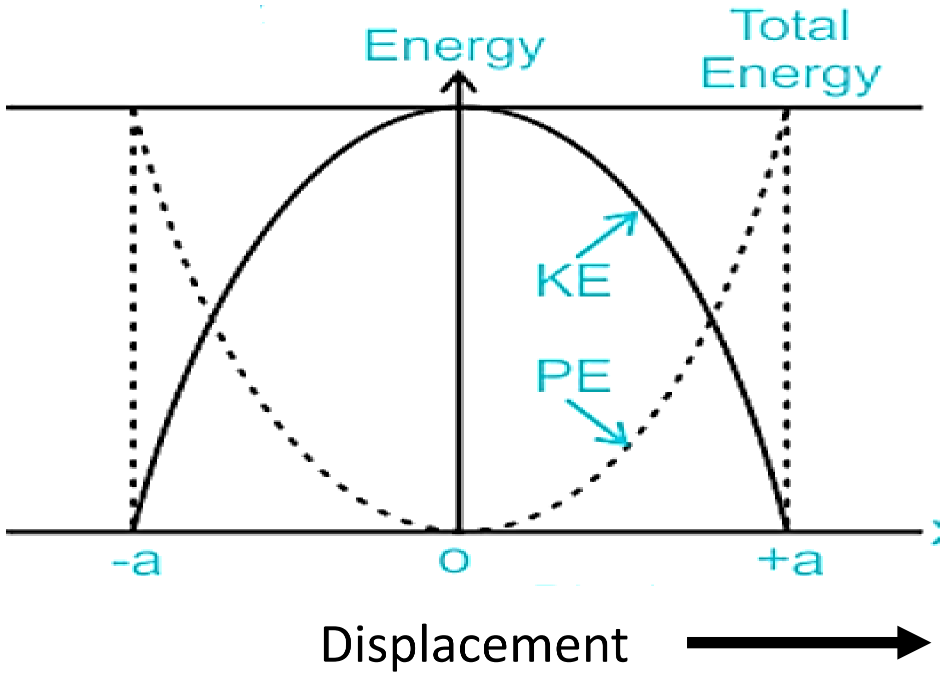


Figure: Energy vs. Displacement graph.

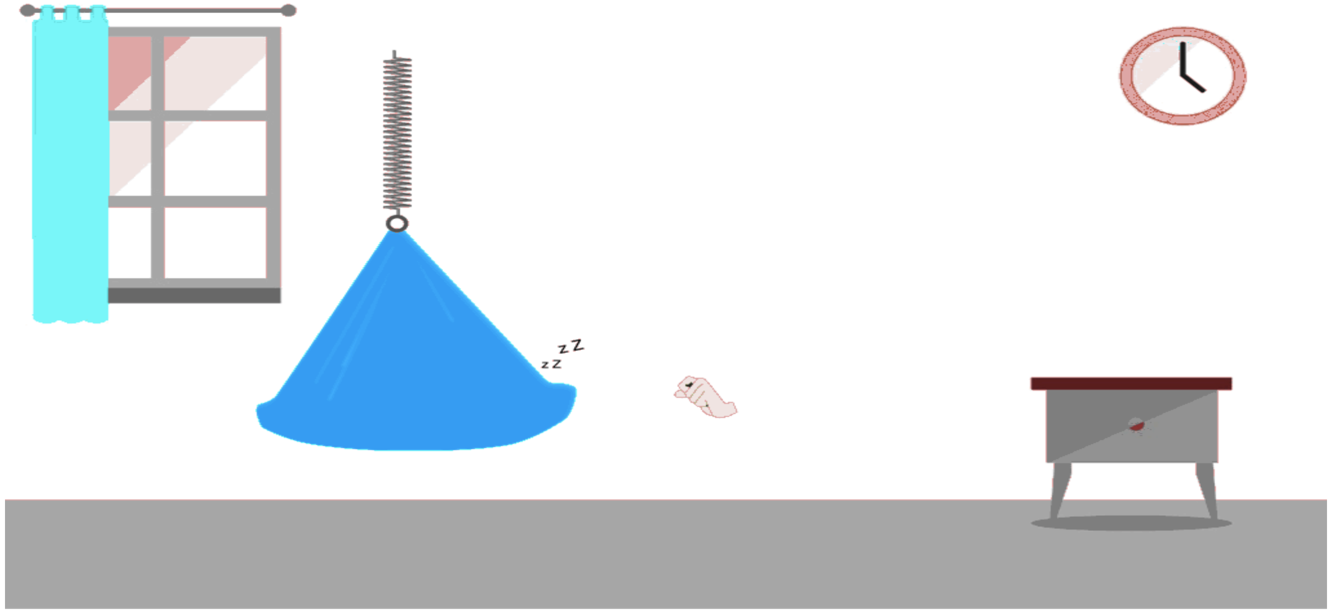
i) If amplitude is maximum (forth & back), the P.E. is maximum and K.E. is minimum.

ii) At equilibrium, K.E. is max. and P.E. is min.

Free Vibration

Free Vibration
Definition:
Free vibration is the type of vibration in which a body, once displaced from its equilibrium position and released, vibrates **with its own natural frequency** without any external periodic force acting on it. Example

Examples:
1) A pendulum displaced and released swings freely.
2) A tuning fork struck and allowed to vibrate in air.



Key Features:
- The system oscillates at its **natural frequency** only.
- Amplitude remains constant if there is **no damping** (ideal case).
- In real systems, amplitude gradually decreases due to damping (air resistance, friction, etc.).

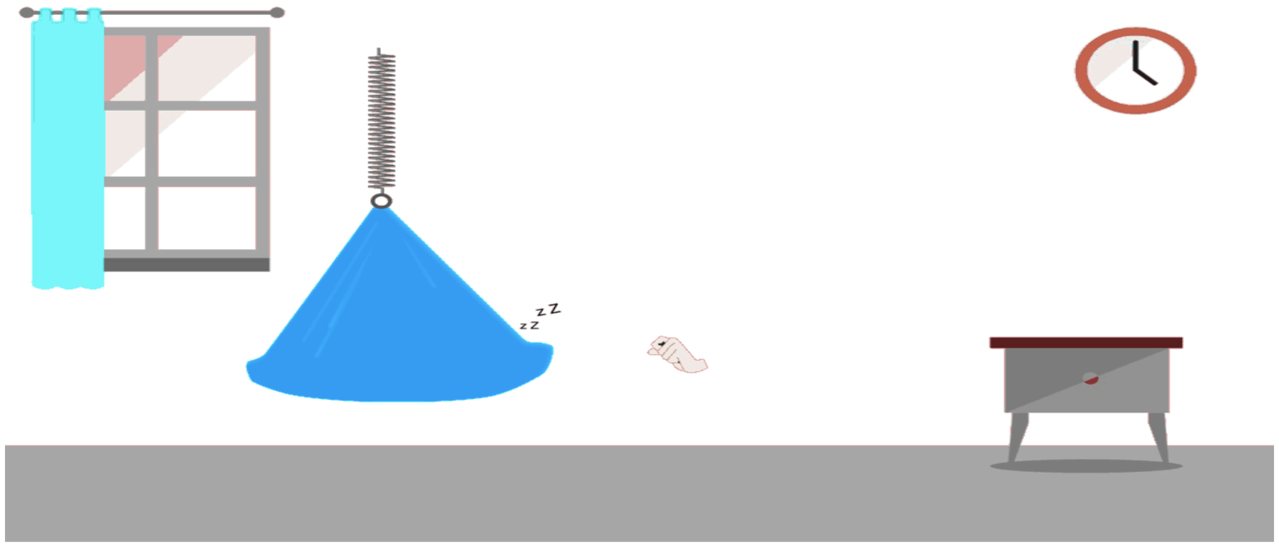
Forced Vibration

Forced vibration:

A **forced vibration** is the type of vibration produced in a body when it is made to vibrate continuously by the action of an **external periodic force**. Example

Example:

The body experiences vibrations due to the periodic vibrations of the vehicle's engine or road bumps.

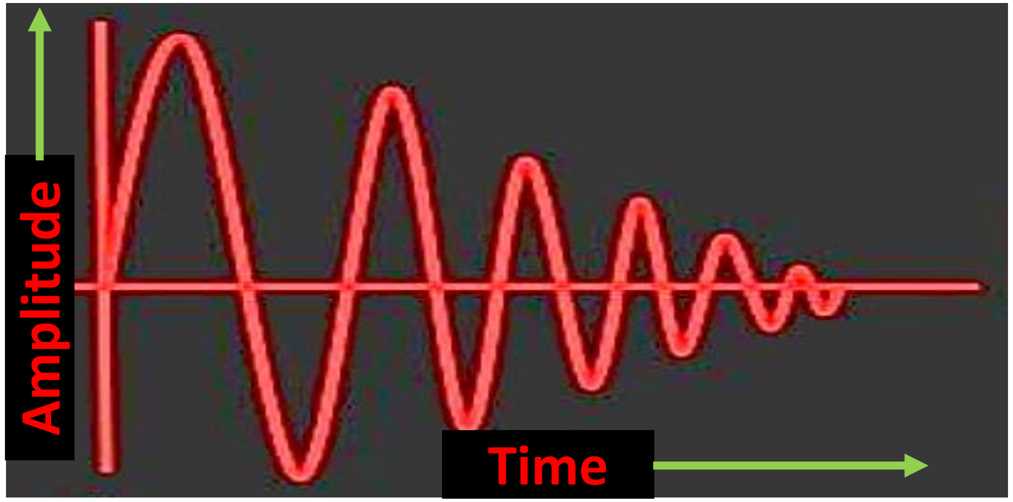


Main Characteristics:

1) The body vibrates **with the frequency of the external force**, (not with its own natural frequency).
2) The **amplitude** depends on how close the external frequency is to the natural frequency and on the damping present.
3) If the external frequency equals the natural frequency, the system may enter **resonance**, where amplitude becomes very large (if damping is small).

Damped & Undamped Oscillations

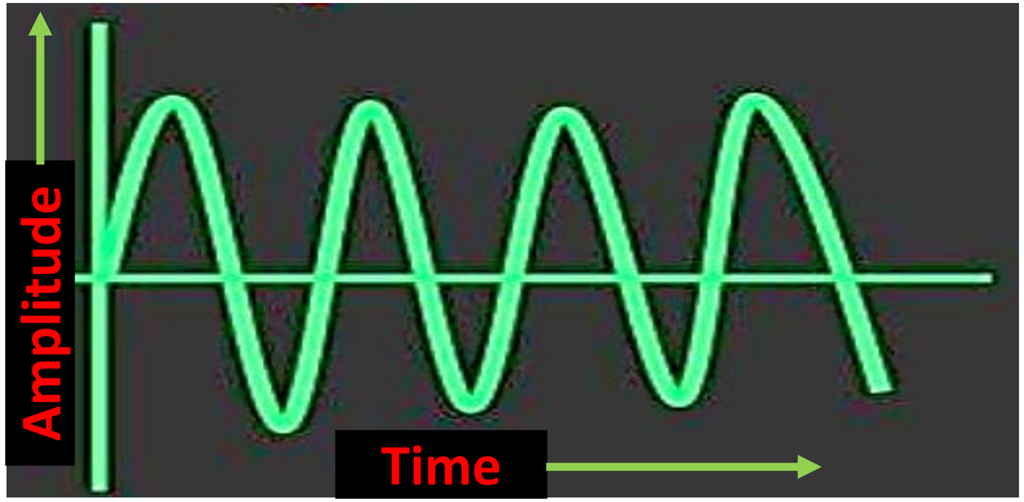
Damped Oscillations



Definition: Oscillations in which the amplitude gradually decreases with time due to energy loss due to friction or resistance.

Example: A pendulum swinging in air, a tuning fork losing sound, or an RLC circuit with resistance.

Undamped Oscillations



Definition: Oscillations in which the amplitude remains constant with time (no loss of energy). No resistive or frictional forces (ideal case) works.

Example: A simple pendulum oscillating in an ideal vacuum (no air resistance).

Mathematical Problem

1) Problem: A particle performs simple harmonic motion (SHM) according to the equation.

$$x = 5\sin\left(4t + \frac{\pi}{6}\right)$$

Where, x is in centimeters and t is in seconds.

Find: **(i) Amplitude, (ii) Angular frequency, (iii) Initial Phase (epoch), (iv) Phase, (v) Time Period, (vi) Frequency, (vi) Displacement after t = 1.2s**

Answer:

The ideal equation for the position of a particle in S.H.M,

$$x = A\sin(\omega t + \varphi) \ldots\ldots\ldots\ldots\ldots\ldots (1)$$

$$x = 5\sin\left(4t + \frac{\pi}{6}\right) \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Now, comparing equations (1) and (2) we get,

(i) Amplitude, A = **5 cm**

(ii) Angular Frequency, $\omega = \mathbf{4\ rad/s}$

(iii) Initial phase (epoch), $\boldsymbol{\varphi_0 = \frac{\pi}{6}}$

(iv) **Phase,** $(\boldsymbol{\omega t} + \boldsymbol{\varphi}) = \left(4t + \frac{\pi}{6}\right)$

(V) Time Period, T = $\frac{2\pi}{\omega} = \frac{2\pi}{4} = \boldsymbol{\frac{\pi}{2}}$

(vi) Frequency, $f = \frac{1}{T} = \frac{1}{\frac{\pi}{2}} = \boldsymbol{\frac{2}{\pi}}$

(vii) Displacement after t = 1.2 seconds,

$$x = 5\sin\left(4t + \frac{\pi}{6}\right)$$

$$= 5\sin\left(4 \times 1.2 + \frac{\pi}{6}\right)$$

$$= \boldsymbol{5\ sin\ (5.3236)}$$

$$= \boldsymbol{-4.08\ cm}$$

Mathematical Problem

2) Problem:

A 0.5 kg mass is attached to a horizontal spring and oscillates with an amplitude of 10 cm. If the spring constant k = 200 N/m,

Find: (i) Angular frequency, (ii) Time period, (iii) Frequency, (iv) Maximum velocity, (v) Maximum acceleration, (vi) velocity & acceleration at t = 0.25 s if it starts from rest at mean position.

Solution: (1) $\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{200}{0.5}} = \sqrt{400} = 20\, rad/s$

(2) T = $\frac{2\pi}{\omega}$ = $\frac{2\pi}{20} = \frac{\pi}{10} = \mathbf{0.314\ sec}$

(3) $f = \frac{1}{T} = \frac{1}{0.314}$

(4) Maximum velocity, $V_{max} = A\omega = 0.1 \times 20 = 2\, m/s$

(5) Maximum acceleration, $a_{\max} = A\omega^2 = 0.1 \times 20^2 = 40\,\mathrm{m/s^2}$

(6) $x = A\sin(\omega t)$

= 0.1 sin (20 × 0.25)

= $-0.0959\, m$

(7) $\mathrm{V} = A\omega$ cos (ωt) = 0.568 m/s

Mathematical Problem

3) Problem: A box of mass 2 kg is connected to the free end of a spring of spring constant 250 N/m. Now a bullet of mass 20 gm with initial velocity of 200 m/s strikes the spring mass. Now find: (i) Time period (T), (ii) Amplitude (A), (iii) Angular frequency (ω) (iv) Equation of displacement.

<u>**Solution:**</u> Let, the mass of the box be M = 2kg
the mass of the bullet be m = 20 gm = 0.020 gm
the initial velocity is $v_0 = 200 m/s$
the initial velocity of the box, $u = 0$
Spring constant, k = 250 N/m

If the final velocity is V then after collision between the bullet and the mass connected to the spring, we can write,

$$(m+M)V = mv_o + Mu$$

$$\text{or, } \boldsymbol{V = \frac{mv_0}{M+m} = \frac{0.02 \times 200}{2+0.02} = 1.98\, m/s}$$

From energy conservation we get,

$$\frac{1}{2}KA^2 = \frac{1}{2}(m+M)V^2$$

$$or, A = \sqrt{\frac{(m+M)V^2}{k}}$$

$$\therefore \boldsymbol{A = 0.178\, m}$$

$$\therefore \boldsymbol{T = 2\pi\sqrt{\frac{m+M}{k}} = 0.565\, sec}$$

$$\therefore \boldsymbol{\omega = \frac{2\pi}{T} = 11.12\, rad/s}$$

$$\therefore \boldsymbol{Equation\ of\ Displacement}$$

$$\boldsymbol{X = A \sin(\omega t + \varphi)}$$

$$\therefore \boldsymbol{X = 0.178 \sin(11.12\, t + \varphi)}$$

Mathematical Problem

4) Problem: The simple harmonic motion of a particle is given by, $\boldsymbol{X = 0.178\, sin\,(11.12\, t +\ \varphi)}$. *Now if the displacement is 2 cm at 0 sec then, Find out (i) The epoch (ii) The frequency (iii) The maximum velocity (iv)Maximum Acceleration at 1m distance* **(v) Phase at t= 3 sec (vi) If the spring constant is 5 N/m, find the total energy.**

(i) Given, $\boldsymbol{X = 0.178\, sin\,(11.12\, t +\ \varphi)}$

or, $0.02 = 0.178 \sin (11.12 \times 0 + \boldsymbol{\varphi})$

or, $\varphi = sin^{-1}\left(\frac{0.02}{0.178}\right)$

or, $\varphi = 6.56^0$

or, $\varphi = \frac{6.56\pi}{180}$

$\therefore \varphi = 0.1145^c$

(ii) The frequency,

$$f = \frac{1}{T} = \frac{1}{\frac{2\pi}{\omega}} = \frac{\omega}{2\pi}$$

$$= \frac{11.12}{2 \times 3.1416}$$

$$= 1.769\ Hz$$

(iii) $v_{max} = \omega A$

$= (11.12 \times 0.178)$

$= 1.979\ ms^{-1}$

(iv) $a_{max} = -\omega^2 x$

or, $a_{max} = -11.12^2 \times 1$

$\therefore\ a_{max} = 123.65\ ms^{-1}$

(v) Phase at t = 3sec

Phase = $(\omega t + \varphi)$

$= (11.12 \times 3 + 0.1145) = 33.4745\ Radian$

(vi) Total Energy, E = $\frac{1}{2}kA^2 = \frac{1}{2} \times 5 \times 0.178^2$

$= 0.07921\ j$

Thank you